

PMARS: A Reproducible Discrete-Event Prototype for Deadline-Aware Satellite PROCESS–STORE–TRANSMIT Scheduling

IASTAM 6.0 — Track 1, Problem 1: Process or Transmit?

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Abstract—The IASTAM 6.0 Track 1 challenge asks a resource-constrained satellite to decide whether each task should be processed onboard, stored, or transmitted. PMARS is a reproducible Python/SimPy discrete-event prototype for evaluating this decision under synthetic energy, storage, compute, memory, thermal, contact-window, deadline, and freshness assumptions. Evaluation V3 compares six schedulers over seven scenarios and twenty matched workload seeds, totaling 840 scheduler runs and 344,970 terminal task outcomes. The primary metric is nominal mission value delivered by deadline divided by all generated nominal value. Rule-based scheduling has the highest observed equal-weight macro-average retention (3.591%), followed by Contact Knapsack (3.501%), EDF (3.152%), Predictive (2.878%), Greedy (2.376%), and Random (2.069%). Predictive is therefore not the overall winner, although it leads in the low-energy scenario. The corrected results show that scheduler performance is regime-dependent. The study reports synthetic simulator evidence only; Raspberry Pi measurements, radio validation, and physical satellite trials remain future work.

Index Terms—satellite edge computing, discrete-event simulation, task scheduling, resource management, reproducible evaluation

I. INTRODUCTION

Satellite instruments generate data while ground contacts, energy, compute capacity, memory, and storage remain time varying. Processing can reduce payload size or create a mission product, but consumes time and energy and may encounter thermal limits. Storage postpones action while occupying finite capacity and reducing freshness. Transmission avoids local processing but requires a sufficiently capable contact before the deadline. PROCESS–STORE–TRANSMIT scheduling is therefore a coupled resource-allocation problem.

The IASTAM 6.0 Track 1, Problem 1 challenge asks a scheduler to choose among immediate onboard processing,

storage for later reconsideration, and transmission to the ground while respecting energy, compute, memory, storage, priority, contact-window, bandwidth, latency, freshness, deadline, and value constraints [1]. PMARS operationalizes these requirements as a controlled single-satellite simulation. Schedulers receive identical generated tasks for each scenario/seed pair, but their decisions create different subsequent resource states. Orbital edge computing similarly treats computation placement as a systems problem coupled to energy and communication opportunities [2].

Research Questions. **RQ1:** How does policy performance vary across energy, communication, workload-burst, critical-task, and thermal-stress regimes? **RQ2:** How does the bounded Predictive heuristic compare with Random, Rule-based, Greedy, EDF, and Contact Knapsack baselines in timely mission-value retention? **RQ3:** What tradeoffs appear among timely value, completion rate, latency, modeled energy, contact use, storage use, and terminal outcomes?

This paper contributes:

- an inspectable PROCESS–STORE–TRANSMIT simulation architecture;
- an outcome-complete task ledger;
- six reproducible scheduler policies;
- seven controlled synthetic regimes;
- a matched-workload evaluation protocol;
- a corrected current-contact packing baseline;
- an exploratory sensitivity screening; and
- a trace-based demonstration of the decision lifecycle.

Reproducibility boundary: Evaluation V3 was generated from source commit 24b8649. The manifest records the scenario and model-profile hashes, dependency versions, seed protocol, and a clean source tree. Evaluation V3 corrects a previously

identified Contact Knapsack capacity-reconstruction issue and reports results from the corrected implementation.

II. CHALLENGE ALIGNMENT AND RELATED WORK

A. Challenge Alignment

PMARS exposes task size, processing demand, memory need, priority, value, deadline, creation time, and freshness decay. Its environment enforces modeled storage admission, energy evolution, processing eligibility, thermal throttling, contact feasibility, and deadlines. A scheduler selects one feasible action at a time, and the simulator records one terminal outcome for every generated task. Evaluation compares timely mission-value retention, timely completion, latency, modeled energy, contact use, storage use, and terminal outcomes. PMARS follows a simulation-first methodology; physical board calibration is reserved for Phase 3 and is described in Section VII.

B. Orbital Computing, Freshness, and Hardware Effects

Denby and Lucia frame orbital edge computing as computation near spaceborne sensors under orbital resource and communication constraints [2]. Trabant uses persistent invocation queues and buffered outputs for intermittent orbital execution [3], while Phoenix studies sunlight-aware placement across satellite and ground resources [4]. PMARS does not reproduce those architectures; its focus is a challenge-oriented simulator and matched scheduler comparison.

Onboard compression can reduce downlink demand, but output quality and downstream utility must remain explicit. Guerrisi *et al.* demonstrate this requirement for Φ -Sat-2 image compression [5]. Freshness also differs from ordinary completion latency: age-of-information research measures the age of the newest delivered source update [6]. PMARS consequently retains nominal deadline-qualified value as its primary metric and treats freshness-adjusted value as secondary. Measurements on commercial onboard computers show interactions among workload, power, and temperature that fixed coefficients cannot fully capture [7]; physical calibration is therefore reserved for future work.

III. SYSTEM MODEL AND IMPLEMENTATION

A. Architecture and Task Lifecycle

PMARS uses SimPy 4.1.2 as a virtual-time environment [8]. A single-satellite, non-preemptive control loop advances arrivals and actions, invokes a scheduler, updates resource models, and records terminal outcomes. Fig. 1 summarizes the evidence flow.

At arrival, a task is admitted only if its input fits in available storage; otherwise, it receives a rejected terminal outcome that remains in the primary-metric denominator. PROCESS occupies the modeled processor and can replace raw data with a processed representation. STORE retains the current representation. TRANSMIT requires a feasible current contact. IDLE advances environmental state. Actions are non-preemptive, while arrivals, deadline expiries, and sunlight boundaries crossed during an action are applied at their simulated timestamps.

The satellite model includes finite battery energy and storage, solar/eclipse cycles, compute and memory eligibility, a single processor, simple thermal evolution and throttling, time-varying contacts, bandwidth, latency, deadlines, and freshness decay. Storage is released when a task becomes terminal. The link model checks one-contact transfer feasibility and does not represent packet retransmission, interference, shared-link contention, or capacity reservation. Arrivals stop at the 3,600-s scenario horizon, after which scheduling drains through the latest generated deadline. All coefficients are configured synthetic assumptions rather than device measurements.

B. Scheduling Policies

The evaluated schedulers are:

- **Random:** seeded selection from currently feasible task-action pairs.
- **Rule-based:** transparent urgency, criticality, priority, value, and current-contact feasibility rules.
- **Greedy:** an immediate-utility heuristic without future projection.
- **Predictive:** a bounded one-step heuristic with a 300-s horizon. It evaluates raw transmission, process-then-transmit, and store-until-contact routes, rejects routes projected to miss the deadline, executes one action, observes the updated state, and replans.
- **EDF:** feasibility-aware earliest-deadline-first scheduling.
- **Contact Knapsack:** a corrected current-contact value-packing reference under the simulator's whole-second duration-bin model. It performs exact 0/1 packing over the remaining current-contact capacity and replans after each scheduling action; it is not a global mission optimizer.

Correction provenance. The historical implementation used dynamic programming over whole-second transfer-duration bins and replanned after each transfer. The Evaluation V2 implementation reconstructed the selected subset only from the exact full contact-capacity bin. It could therefore select a 10-s value-1 task over a 9-s value-100 task in a 10-s contact, and could return IDLE for three feasible 3-s tasks in the same contact. The corrected implementation selects the highest-value reachable capacity. Dedicated tests cover both counterexamples, and the corrected implementation passed 27 regression tests in the audited environment.

For task i with projected delivery time t_d , Predictive uses the normalized value term

$$f_i(t_d) = \frac{v_i(t_d)}{\max(v_i, 1.0)}, \quad (1)$$

where $v_i(t_d)$ is freshness-adjusted value and v_i is nominal value. The fixed implementation score is

$$S_i = 4.0f_i(t_d) + 1.5p_i + 0.5\ell_i + 0.75b_i + 0.5I_{\text{critical},i} - 0.7e_i - 0.3w_i, \quad (2)$$

where p_i is normalized priority, ℓ_i is normalized deadline slack, b_i is normalized storage relief, e_i is projected energy normalized by current energy, and w_i is waiting time normalized by the planning horizon. $I_{\text{critical},i}$ identifies a critical task. Predictive

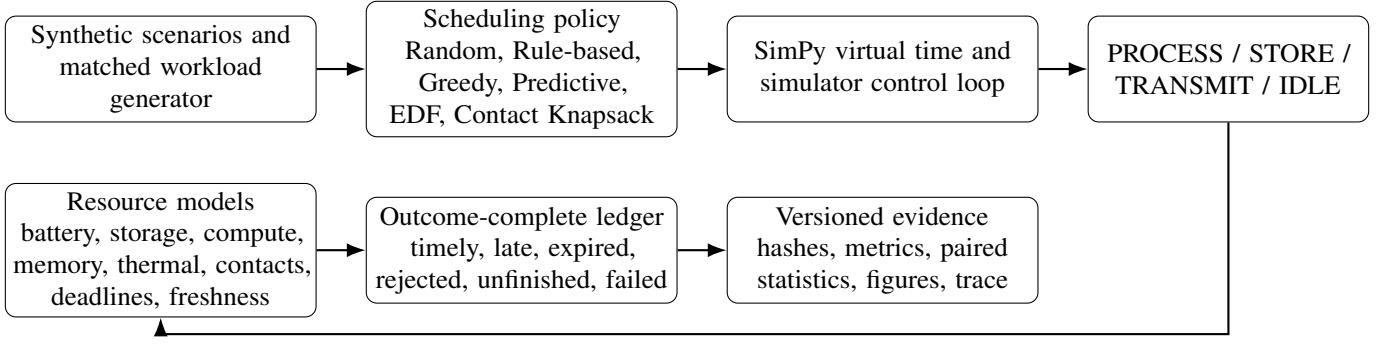


Fig. 1. PMARS architecture and evidence flow. All schedulers receive the same generated workload for a scenario/seed pair; their actions then produce different resource states.

also checks future-contact feasibility when forming candidate routes. It is not machine learning, not trained, not a global optimizer, not an optimality proof, and does not reserve future-contact capacity. These are fixed synthetic implementation parameters, not learned or hardware-calibrated values.

IV. EVALUATION METHODOLOGY

A. Scenarios and Matched Runs

Evaluation V3 contains seven scenarios: normal operation, critical-task stress, low energy, poor link, sustained task burst, combined energy-link stress, and thermal-burst stress. The matrix is

$$7 \text{ scenarios} \times 6 \text{ schedulers} \times 20 \text{ seeds} = 840 \text{ runs.}$$

Workload seeds 100–119 are matched across schedulers. A scenario/seed workload is generated once and reused for every scheduler; Random uses a separate reproducible policy seed, while the other schedulers are deterministic for the supplied workload and state. This controls input tasks but does not force identical subsequent states.

B. Metrics and Integrity Checks

The primary metric is timely mission-value retention:

$$RV = 100 \frac{\sum_i v_i \mathbf{1}[o_i = \text{timely}]}{\sum_i v_i}, \quad (3)$$

where the denominator contains all generated nominal mission value, including rejected, expired, late, unfinished, and failed tasks. Freshness affects the secondary freshness-adjusted metric and the Predictive route score; the primary timely-value-retention metric uses nominal mission value and does not apply freshness decay. Scenario-policy cells report the mean and two-sided 95% Student's t interval over twenty matched seeds. Predictive-versus-baseline comparisons are matched-seed, exploratory, nominal, and not adjusted for multiple comparisons.

The exported ledger contains 344,970 scheduler-run task outcomes and satisfies

$$N_{\text{generated}} = N_{\text{timely}} + N_{\text{late}} + N_{\text{expired}} + N_{\text{rejected}} + N_{\text{unfinished}} + N_{\text{failed}} \quad (4)$$

for every run. Workload hashes verify cross-scheduler task identity for each scenario/seed pair.

V. EVALUATION V3 RESULTS

A. Timely Mission-Value Retention

Table I and Fig. 2 summarize the primary result. Bold marks the highest observed mean in each scenario.

The macro-average ranking is Rule-based (3.591%), Contact Knapsack (3.501%), EDF (3.152%), Predictive (2.878%), Greedy (2.376%), and Random (2.069%). Predictive is not the overall winner. Rule-based scheduling leads Critical (5.741%), Energy-link stress (0.427%), Normal (13.408%), Task burst (2.626%), and Thermal burst stress (2.468%). Predictive leads Low energy (0.516%), while Contact Knapsack leads Poor link (0.382%). No scheduler dominates all regimes.

The exploratory paired results reinforce this regime dependence. In Low energy, Predictive exceeds Greedy by 0.451 percentage points [0.320, 0.582], Rule-based by 0.420 [0.284, 0.556], EDF by 0.445 [0.314, 0.576], and Contact Knapsack by 0.445 [0.314, 0.576]. Conversely, Predictive trails Rule-based in Normal by 4.161 points [−4.821, −3.501] and in Critical by 1.022 points [−1.334, −0.710]. These intervals are exploratory, nominal, and not multiplicity-adjusted.

B. Operational Tradeoffs

Table II reports selected operational metrics for normal operation, energy scarcity, and sustained arrivals; Fig. 3 shows the corresponding value-energy tradeoff. The selection includes all schedulers and illustrates that retention rankings cannot be inferred from a single resource metric.

In Low energy, all schedulers consume the same mean modeled energy to the displayed precision, yet timely value and contact use differ substantially. In Task burst, Predictive has the shortest mean timely latency and the highest contact use, while Rule-based retains slightly more timely mission value. These observations are descriptive and do not identify a universal bottleneck.

C. Terminal Outcomes

The V3 ledger contains 27,737 timely completions, 303,590 expiries, and 13,643 admission rejections, with zero late, unfinished, or failed outcomes. Fig. 4 summarizes their composition by scheduler. The high expiry fraction occurs under the deliberately demanding configured synthetic regimes. The aggregate

TABLE I
EVALUATION V3 TIMELY MISSION-VALUE RETENTION (%). EACH SCENARIO CELL IS MEAN $\pm$ THE TWO-SIDED 95% STUDENT'S t INTERVAL HALF-WIDTH OVER TWENTY MATCHED SEEDS.

Scenario	Random	Rule-based	Greedy	Predictive	EDF	Contact Knapsack
Critical	3.948 $\pm$ 0.281	5.741 $\pm$ 0.295	4.284 $\pm$ 0.246	4.719 $\pm$ 0.264	5.606 $\pm$ 0.297	5.628 $\pm$ 0.309
Energy-link stress	0.253 $\pm$ 0.052	0.427 $\pm$ 0.076	0.203 $\pm$ 0.024	0.337 $\pm$ 0.044	0.285 $\pm$ 0.035	0.293 $\pm$ 0.037
Low energy	0.127 $\pm$ 0.085	0.096 $\pm$ 0.081	0.065 $\pm$ 0.058	0.516 $\pm$ 0.137	0.071 $\pm$ 0.070	0.071 $\pm$ 0.070
Normal	7.469 $\pm$ 0.476	13.408 $\pm$ 0.707	8.934 $\pm$ 0.450	9.247 $\pm$ 0.301	11.425 $\pm$ 0.454	13.228 $\pm$ 0.719
Poor link	0.277 $\pm$ 0.060	0.368 $\pm$ 0.065	0.375 $\pm$ 0.067	0.370 $\pm$ 0.066	0.380 $\pm$ 0.069	0.382 $\pm$ 0.067
Task burst	1.398 $\pm$ 0.074	2.626 $\pm$ 0.077	1.427 $\pm$ 0.080	2.610 $\pm$ 0.073	2.211 $\pm$ 0.068	2.527 $\pm$ 0.077
Thermal burst stress	1.014 $\pm$ 0.098	2.468 $\pm$ 0.091	1.346 $\pm$ 0.106	2.346 $\pm$ 0.073	2.084 $\pm$ 0.061	2.377 $\pm$ 0.068
Macro-average	2.069	3.591	2.376	2.878	3.152	3.501

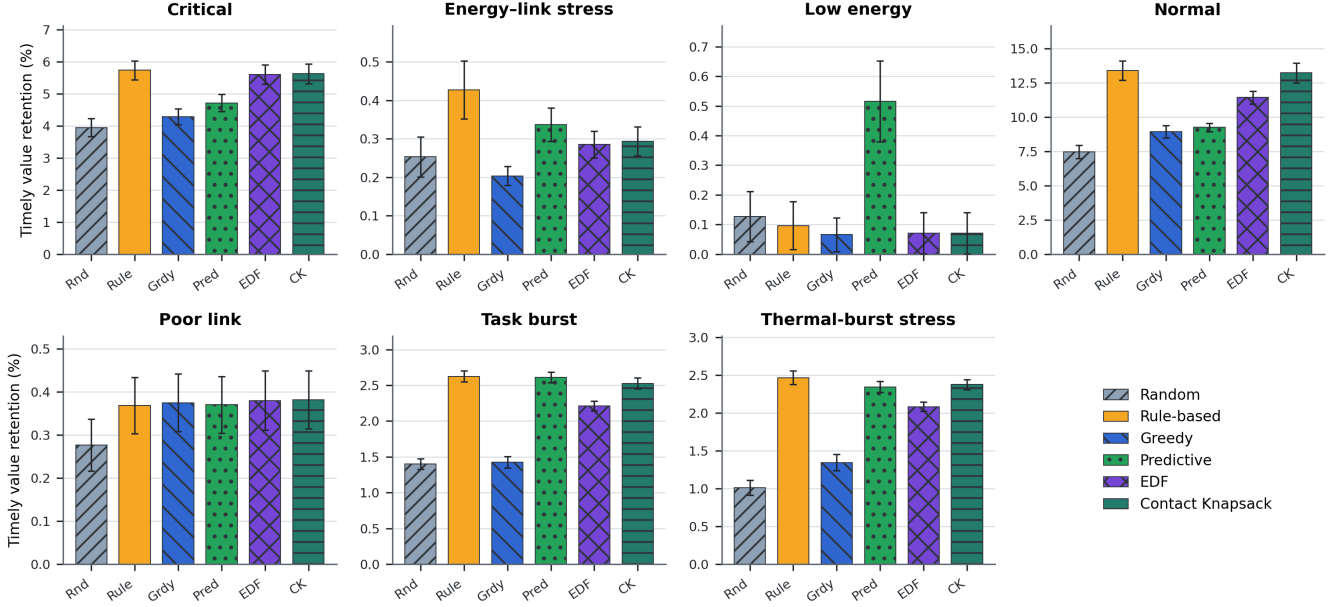


Fig. 2. Evaluation V3 deadline-qualified mission-value retention. Bars are scenario-policy means and whiskers are two-sided 95% Student's t intervals. Each scenario uses an independent zero-based vertical range selected for readability; therefore, vertical scales are not directly comparable across panels. Short labels are Rnd (Random), Rule (Rule-based), Grdy (Greedy), Pred (Predictive), EDF, and CK (Contact Knapsack).

evaluation alone does not identify which individual resource constraint principally drives expiry; exploratory parameter screening provides limited directional evidence.

D. Illustrative Lifecycle Trace

Table III and Fig. 5 summarize Normal / Predictive / seed 100 for telemetry task 100-1. The trace demonstrates arrival, queue admission, PROCESS, STORE/wait behavior, contact use, TRANSMIT, and the terminal outcome. The event record can also expose energy, storage, link state, deadline slack, and scheduler rationale. This is an illustrative simulator trace, not aggregate performance evidence and not a hardware measurement.

E. Exploratory Sensitivity Screening

Table IV reports a directional screening that varies six parameters independently across Normal, Low energy, Poor link, and Task burst. It covers all six schedulers at multipliers

0.75 $\times$, 1.00 $\times$, and 1.25 $\times$ using one matched workload seed (seed 100), for $4 \times 6 \times 6 \times 3 = 432$ screening runs.

The screening suggests that communication opportunity and arrival pressure have larger directional effects on absolute retention than the tested initial energy or storage multipliers. Because each condition uses one workload seed and factors are varied independently, these results are exploratory screening evidence rather than multi-seed inferential conclusions. They do not establish causality or statistical significance and do not constitute a complete sensitivity analysis.

VI. LIMITATIONS

Evaluation V3 is a reproducible synthetic simulator study. Workloads, contacts, mission values, energy, processing duration, memory eligibility, bandwidth, latency, freshness, and thermal parameters are configured assumptions. The study uses no physical orbit traces, spacecraft measurements, Raspberry Pi measurements, radio measurements, or flight trials. The link

TABLE II
SELECTED EVALUATION V3 OPERATIONAL MEANS. POLICY ORDER MATCHES TABLE I.

Scenario	Metric	Random	Rule-based	Greedy	Predictive	EDF	Contact Knapsack
Normal	Modeled energy (J)	87,738	97,709	96,527	88,500	97,565	98,046
	Timely latency (s)	195.5	198.9	157.4	121.6	254.5	227.8
	Contact use (%)	38.9	52.6	44.7	44.9	51.6	52.8
	Mean storage (%)	24.3	23.7	23.3	24.8	24.2	24.0
Low energy	Modeled energy (J)	35,188	35,188	35,188	35,188	35,188	35,188
	Timely latency (s)	203.5	92.7	57.6	173.4	99.4	98.8
	Contact use (%)	1.4	0.8	0.5	5.1	0.6	0.6
	Mean storage (%)	26.5	26.2	26.1	27.6	26.1	26.1
Task burst	Modeled energy (J)	91,180	110,554	107,280	93,942	110,733	110,588
	Timely latency (s)	139.5	164.5	49.9	34.1	212.3	166.3
	Contact use (%)	38.2	52.7	34.7	55.5	51.0	50.9
	Mean storage (%)	81.0	80.5	80.6	81.0	80.7	80.4

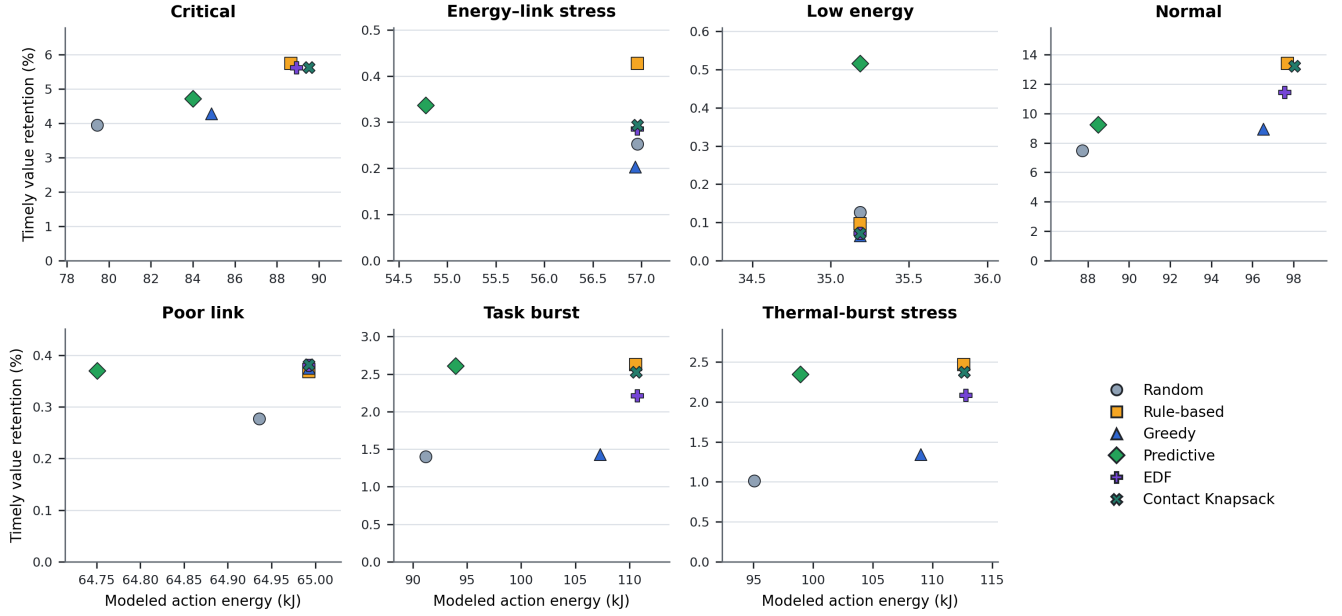


Fig. 3. Evaluation V3 value-energy tradeoff. Each point is a scenario-scheduler mean; panel ranges are scenario-specific and policy identity is encoded by color and marker rather than overlapping text labels. Energy is gross modeled action energy and is not measured hardware power, board energy, or spacecraft energy.

TABLE III
ILLUSTRATIVE EVALUATION V3 LIFECYCLE FOR TASK 100-1; DEADLINE 369.294 s.

Event	Time (s)	Evidence
Arrival	30.440	Task admitted to the queue.
PROCESS	32.440	Processed representation available.
complete		
STORE waiting	32.440–102.440	Fourteen wait cycles precede contact use.
TRANSMIT	105.576	Delivered before the deadline.
complete		
Terminal outcome	105.576	Completed / timely.

TABLE IV
EXPLORATORY SENSITIVITY SCREENING. VALUES ARE MEAN TIMELY MISSION-VALUE RETENTION ACROSS SCREENING ROWS.

Parameter	0.75×	1.00×	1.25×
Available initial energy	3.106	3.103	3.119
Contact bandwidth	2.328	3.103	3.917
Contact-window duration	2.427	3.103	3.616
Interarrival time	2.458	3.103	3.925
Deadline range	3.087	3.103	3.203
Storage capacity	3.116	3.103	3.095

model does not represent interference, packet-level retransmission, cross-contact continuation, or shared future-capacity reservation. The scheduler is non-preemptive and models one satellite. The configured thermal ceiling exercises simulator

throttling only and is not evidence of physical thermal behavior. Consequently, the results support controlled policy comparison and regression analysis, not claims of flight readiness or calibrated mission performance.

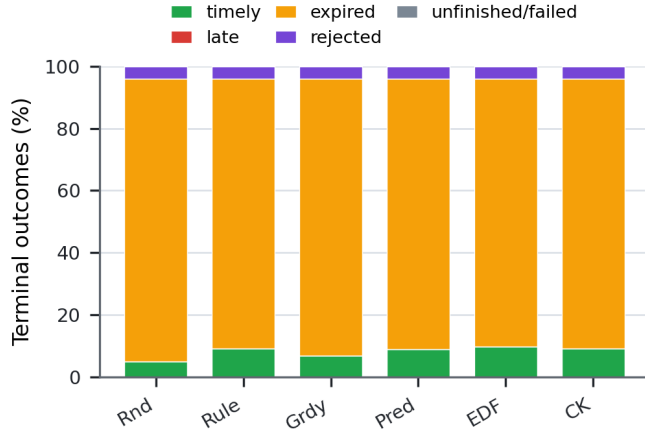


Fig. 4. Evaluation V3 terminal-outcome composition aggregated across the seven scenarios. Scheduler order is Random, Rule-based, Greedy, Predictive, EDF, and Contact Knapsack. Categories are timely, late, expired, rejected, and unfinished/failed; every stacked bar is normalized to sum to 100%.

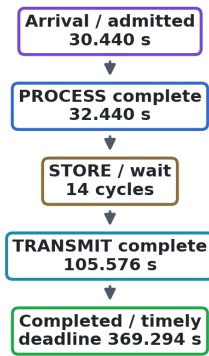


Fig. 5. Illustrative V3 simulator lifecycle for Normal / Predictive / seed 100 / task 100-1. Completion at 105.576 s precedes the 369.294-s deadline; this is not aggregate or hardware evidence.

VII. FUTURE WORK / PHASE 3 HARDWARE VALIDATION

No hardware measurements are reported in this Phase-2 evaluation. Phase 3 will evaluate representative telemetry transforms, image compression, a documented inference or science workload, and idle operation on a Raspberry Pi 5. The planned platform has 8 GB RAM and active cooling and records exact board, firmware, operating-system, kernel, storage, and software versions. Each configuration will include warm-up followed by at least thirty retained trials.

The protocol will record wall-clock duration, input and output size, input-side energy, average and peak power, peak resident memory, board and ambient temperature, CPU frequency, throttling flags, and failures. Communication will be measured separately using controlled Ethernet or Wi-Fi contact/radio emulation, with throughput, latency, transfer energy, contact timing, retransmissions, and failures recorded. Such experiments will be described as emulation unless flight-

like radio hardware is actually tested.

A measured model profile will be created only after raw runs, instrument metadata, derivation scripts, hashes, units, uncertainty, and failure summaries are archived. Simulation-versus-board comparisons will then report duration, energy, memory, temperature, output size, outcome agreement, and prediction error under a new evaluation version. No Raspberry Pi measurement, hardware calibration, radio validation, or physical satellite result is claimed in this paper.

An adaptive meta-scheduler is a natural next direction. It could select among Rule-based, Predictive, EDF, and contact-packing strategies according to observed battery state, contact availability, queue pressure, and thermal state. Such a policy should be evaluated on new held-out workloads rather than tuned on the present evaluation; it has not been implemented or evaluated here.

VIII. CONCLUSION

PMARS is a reproducible discrete-event prototype for deadline-aware PROCESS-STORE-TRANSMIT scheduling. Evaluation V3 contains 840 runs across seven synthetic scenarios, six schedulers, and twenty matched seeds, with 344,970 terminal outcomes. Rule-based scheduling has the highest observed macro-average timely mission-value retention (3.591%), corrected Contact Knapsack is second (3.501%), and Predictive records 2.878%. Predictive leads Low energy, Contact Knapsack leads Poor link, and Rule-based scheduling leads the other five scenarios; no scheduler dominates every regime. The exploratory screening suggests larger directional effects from communication opportunity and arrival pressure within the tested perturbations, without supporting causal or inferential claims. Hardware calibration and adaptive policy selection remain future research directions.

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